

ChE 382: Unit Operations Laboratory

HUMIDIFICATION

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Introduction

In every process involving mass transfer, heat must also be transferred. When a component is transferred from a gas phase to solution in a liquid phase, the latent heat associated with the condensation is involved. When a component is transferred from solution in one solvent to solution in a second, as in liquid-liquid extraction, the difference between the heats of solution of the solute in the two solvents is involved. Similar heat effects are present in distillation, adsorption, leaching, and drying. In many cases, the heat effect is small due to large thermal mass of the contacting phases. This is often the case in distillation, liquid-liquid extraction, gas absorption in dilute solutions, and leaching. In other processes, particularly where there is a net transfer of mass from the gas phase to a condensed phase (or vice versa), the heat-transfer rate is important. It can limit the rate at which mass can be transferred. In other processes, such as boiling, condensation, evaporation and crystallization, mass and heat transfer occur simultaneously in large amounts, however the rate of mass transfer is solely determined by the rate of heat input from an external source.

In the group of processes where both heat transfer and mass transfer affect the rate, humidification and dehumidification are the simplest and most widely used application. Only two components are involved: air and water. This system has been extensively studied by chemical engineers due to its practical importance. There are numerous models and correlations which have been developed to help the practicing engineer analyze experimental data and to rationally design equipment (e.g., cooling towers, drying processes).

In addition, the theory of simultaneous heat and mass transfer can also be applied to the analysis of other processes. An example is chemical reactors. In these processes, there is mass transfer due to the chemical reaction which is accompanied by a large heat of reaction.

This experiment uses humidification to generate a set of experimental data which can be used with existing correlations to test the accuracy of the data.

Theory

The transfer of heat and mass in most unit operations is usually described in terms of transport coefficients because the flow patterns in these processes are too complex to analyze in detail. The flux equations in terms of transport coefficients are:

$$q = h \cdot \Delta T \quad (1)$$

$$N_A = k_c \cdot \Delta c_A \quad (2)$$

Where:

- q and N_A are the heat and mass fluxes, respectively (energy or moles/area-time)
- ΔT is a characteristic temperature difference
- Δc_A is a characteristic concentration difference
- h and k_c are the heat and mass transfer coefficients.

Equations (1) and (2) define the transport coefficients as the ratio of the flux to a linear driving force. Note that the mass transfer coefficient will take on different numerical values depending on the units used for the driving force term; for example, concentration, mole fraction or ratio, partial pressure, mass fraction. One must select a convenient set of units consistent with the experimental measurements. There are also numerous correlations available for the mass and heat transfer coefficients as a function of the system geometry and the typical process parameters and physical properties of the components. Charts are available which correlate the heat and mass transfer without directly going through the use of heat and mass transfer coefficients. The choice of a wetted-wall column is particularly attractive since extensive analysis has been done for this simple and well-defined geometry.

In order to use these equations to calculate a flux or alternatively, to obtain the concentration or temperature of the fluid, specific numerical values for the coefficients must be available. These are usually obtained from appropriate experimental measurements made with an apparatus having a flow geometry similar to that of the desired system. Correlations are usually reported in terms of appropriate dimensionless groups, and for some well-specified bulk average concentration or temperature of the fluid. An example of a correlation for the gas-phase mass transfer coefficient inside a circular pipe is:

$$Sh = 0.023 \cdot Re^{0.83} \cdot Sc^{0.44} \quad (3)$$

Where:

- Re is the Reynolds number based on the pipe diameter
- Sc is the Schmidt number
- Sh is the Sherwood number defined as: $Sh = k_C \cdot L / D_{AB}$ (4)

Where:

k_C is the mass transfer coefficient for a concentration driving force

L is a characteristic length scale defined for the system under study

D_{AB} is the diffusion coefficient for the fluid through which the mass transport occurs (in this experiment, the air).

Mass Transfer. The mass transfer coefficient defined by (4) is for low mass transfer rates and dilute concentrations of the transferring component. If these restrictions are not fulfilled, the mass transfer coefficient obtained from (3) cannot be used directly in (2) without first correcting the coefficient for the rate of mass transfer and for situations where the diffusion is not equimolar counter diffusion.

In general, the mass transfer coefficient is not affected by the simultaneous transfer of heat, except in as much as D_{AB} and the dimensionless groups are functions of temperature in a non-isothermal system. However, the heat transfer coefficient will be influenced by mass transfer since the transferring species carries with it some energy. The flux of the diffusing component in a fluid may increase or decrease the thermal conductivity of the fluid (depending on the direction of the mass transfer relative to heat transfer) with respect to that of a pure fluid. This correction factor, however, is small in many applications involving humidification systems and can usually be neglected. Therefore, for situations where the mass flux is small, the diffusing component is present in dilute concentrations, and the thermal conductivity correction term may be neglected, the heat and mass transfer coefficients obtained from appropriate correlations may be used directly in equations (1) and (2) to calculate fluxes.

The application of equations (2) and (3) for calculating the transfer of heat and mass inside the column requires that conservation equations for mass and energy be written over a differential height element since the transport coefficients are defined in terms of the average cross-sectional temperature and composition of the fluid, which both change continuously in the axial direction. A mass balance on the water component of the fluid over the differential height element of the column is shown schematically in Figure 1 and as the differential:

$$d(Gy) = k_c c (y_i - y) dA \quad (5)$$

where:

- y_i and y refer to the mole fraction of water at the interface and in the gas, respectively
- G is the total gas flowrate (moles/time)
- c is the molar concentration of the gas, and dA is the area of the differential column height ($=Pdz$), where P is the perimeter of the tube and dz is a differential height)
- $d(Gy)$ is the differential change in the moles of water in the gas over the corresponding differential column height and is equal to the rate of mass transfer at the fluid interface.

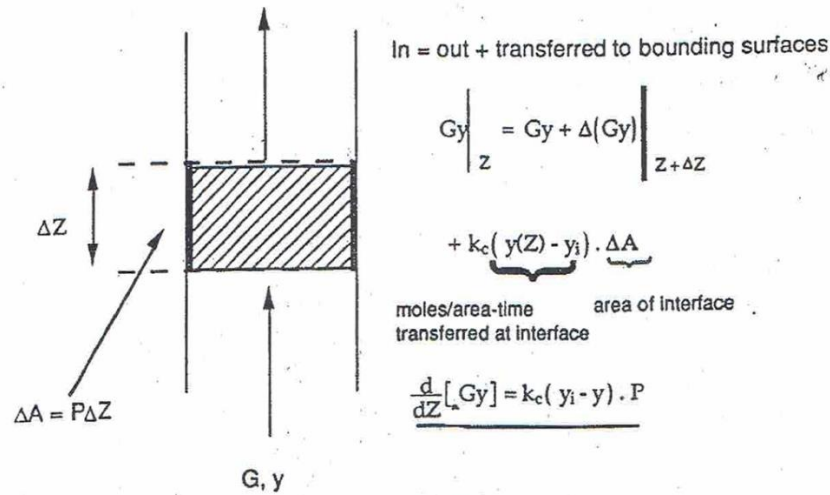


Figure 1. Schematic of a differential element in the column.

The equation is written in terms of the mass transfer coefficient, k_c , defined by equation (2). This is valid provided the transferring species is present at low concentrations and the mass transfer rate is low. If these conditions are not met, the bulk flow term must be included in the definition of the mass transfer coefficient, since this process does not involve equimolar counterdiffusion of the transferring species, and the mass transfer coefficient must be corrected for the high mass transfer rates. In humidification calculations, one frequently expresses the concentration of the vapor (water) in the gas (air) in terms of the mass ratio rather than mole fraction. One can define:

$$Y' = \frac{\text{mass water}}{\text{mass dry gas}} \quad \text{As:} \quad Y' = \left\{ \frac{y}{(1-y)} \right\} \left(\frac{MW_{H_2O}}{MW_{Air}} \right) \quad (6)$$

Where MW_{H_2O} refers to the molecular weight of component y (water) and MW_{Air} is the molecular weight of the air.

Furthermore, the quantity $d(\text{Gy})$ can be written as;

$$\left\{ \frac{G'_S}{MW_{Air}} \right\} d(Y')$$

Where G'_S is the mass flowrate of dry air. The differential mass balance equation for water is now;

$$G'_S d(Y') = k_Y (Y'_i - Y') P dz \quad (7)$$

Where k_Y is a mass transfer coefficient defined in terms of a mass ratio driving force. The mass transfer coefficient, k_Y , rather than k_c , is often used in humidification problems in industry. It is related to the concentration driving forces by the equation;

$$k_Y = k_c (MW_{Air}) c (1 - y)_M \quad (8)$$

Where $(1-y)_M$ is the log mean average composition of the air over the column length. It is defined as:

$$(1 - y)_M = \frac{\{y_1 - y_2\}}{\ln \left\{ \frac{(1-y)_2}{(1-y)_1} \right\}} \quad (9)$$

Where 1 and 2 refer to the compositions at the inlet and outlet of the column.

Heat Transfer. A heat balance over the over a differential height elemet is:

$$d(GC_p T) = h (T_i - T) dA \quad (10)$$

Where:

- C_p is the molar heat capacity of the gas
- T_i is the temperature at the interface and T is the gas temperature.

In humidification problems, the left hand side of equation 10 is usually written as:

$$G'_S C'_S dT$$

Where G'_S is the mass flowrate of dry air and C'_S is the humid heat or the heat capacity of the gas and its accompanying vapor per mass of dry air:

$$C'_S = C'_{Air} + Y' C'_{H2O} \quad (11)$$

Here, C'_{Air} is the heat capacity of the dry air and C'_{H2O} is the heat capacity of the water vapor. Note that C'_S is a function of Y' , the vapor content of the gas. Provided that Y' does not vary appreciably over the length of the column, C'_S is essentially constant.

Integration of equations (7) an (10) over the column height will yield an equation relating the height to the inlet and outlet gas (air) temperatures and compositions, a convenient form for the design of humidification/dehumidification equipment. The integration is straightforward once it is recognized that Y'_i and T_i are constants for the recirculated wetted wall column, and hence any coupling between the two equations is neglected.

For humidification problems, in general, the equations are not strongly coupled and may be solved independently of each other. The results are the equations:

$$\ln \left\{ \frac{Y'_i - Y'_1}{Y'_i - Y'_2} \right\} = \frac{k_Y P z}{G_S} \quad (12)$$

$$\ln \left\{ \frac{T_i - T_1}{T_i - T_2} \right\} = \frac{h P z}{G_S C_S} \quad (13)$$

Where Y'_1 , T_1 , and Y'_2 , T_2 , are the inlet and outlet compositions and temperatures of the gas and its vapor, respectively. The left hand side of equation 12 is usually referred to as the number of transfer units, NTU (or N_{tG}), which is equal to the number of times the average driving force for mass transfer ($Y'_i - Y'$) is divided into the difference in composition between the top and bottom of the column. It is a measure of the number of discrete equilibrium stages necessary to achieve the desired mass transfer. The traditional approach is to describe the column height as the product of a HTU (height of transfer unit) and NTU (number of transfer units). The HTU can be thought of as describing the efficiency of contacting and the NTU as the difficulty of separation.

The theory of humidification is described in the references [1-4] and also numerous other sources such as texts on mechanical engineering and specialized books on topics such as cooling tower design and air conditioning.

Experimental Details and Procedure

We will study the transport of both mass and heat in a wetted wall column and obtain experimental values for the heat and mass transfer coefficients. The measured values for the transport coefficients will be plotted as a function of the Reynolds number and compared to the correlation given in the Theory section (Equation 3). A wetted-wall column is shown schematically in Figure 2. Air enters at the bottom of the column and flows upward countercurrent to the water. Water is allowed to flow down the inside wall of the circular tube. The flowrate of water is sufficiently high that a thin film of water covers the entire surface of the tube. The water is recirculated such that the liquid temperature is essentially constant throughout the column. (This is the same as the “well-mixed” assumption in a CSTR). Heat loss from the outside of the tube and from the water recirculation and distribution lines can be assumed to be negligible so that the operations inside of the column take place adiabatically. The water temperature at steady state operation will correspond to the adiabatic saturation temperature of the incoming gas, i.e. the temperature to which the incoming gas will be cooled if it is allowed to become saturated adiabatically. The adiabatic saturation temperature can be obtained from a psychrometric chart provided the temperature and humidity of the incoming gas is specified.

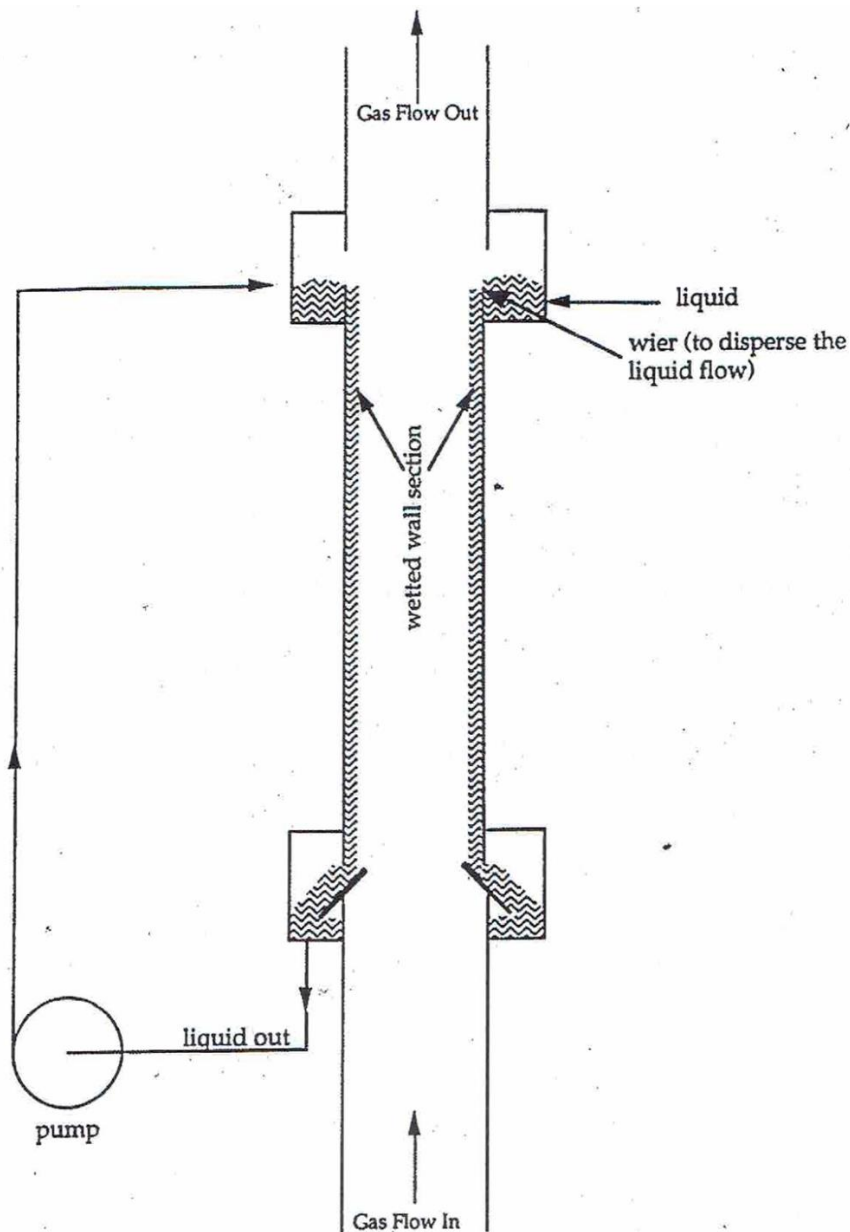


Figure 2. Schematic of a generalized wetted wall column.

Humidification Unit. The wetted-wall humidification unit consists of a 2-inch diameter Plexiglas column 4 feet in length connected to an air flow and water feed system as shown in Figure 3. The air comes from the house supply and the flowrate is determined by a rotameter; the air enters the bottom of the column. The water is pumped from holding tank via a small centrifugal pump and the flowrate is measured using a rotameter; the water enters the top of the column. A weir is used to distribute the water to the sides of the top of the column. Thus, the air and water flow counter-currently.

The temperature of the incoming air is measured with a thermocouple. The temperature of the air exiting the humidification column is measured using a wet-bulb thermometer (top of unit) or an electronic hygrometer. The system run without water can be used to determine the humidity of the incoming air.

The water temperature is measured at the pump outlet and in the holding tank using thermocouples. Three additional thermocouples are mounted on the column wall to measure the temperature of the water flowing down the tube.

The system has a heat exchanger to allow heating of the incoming air. This can be used to obtain data at different rates of humidification.

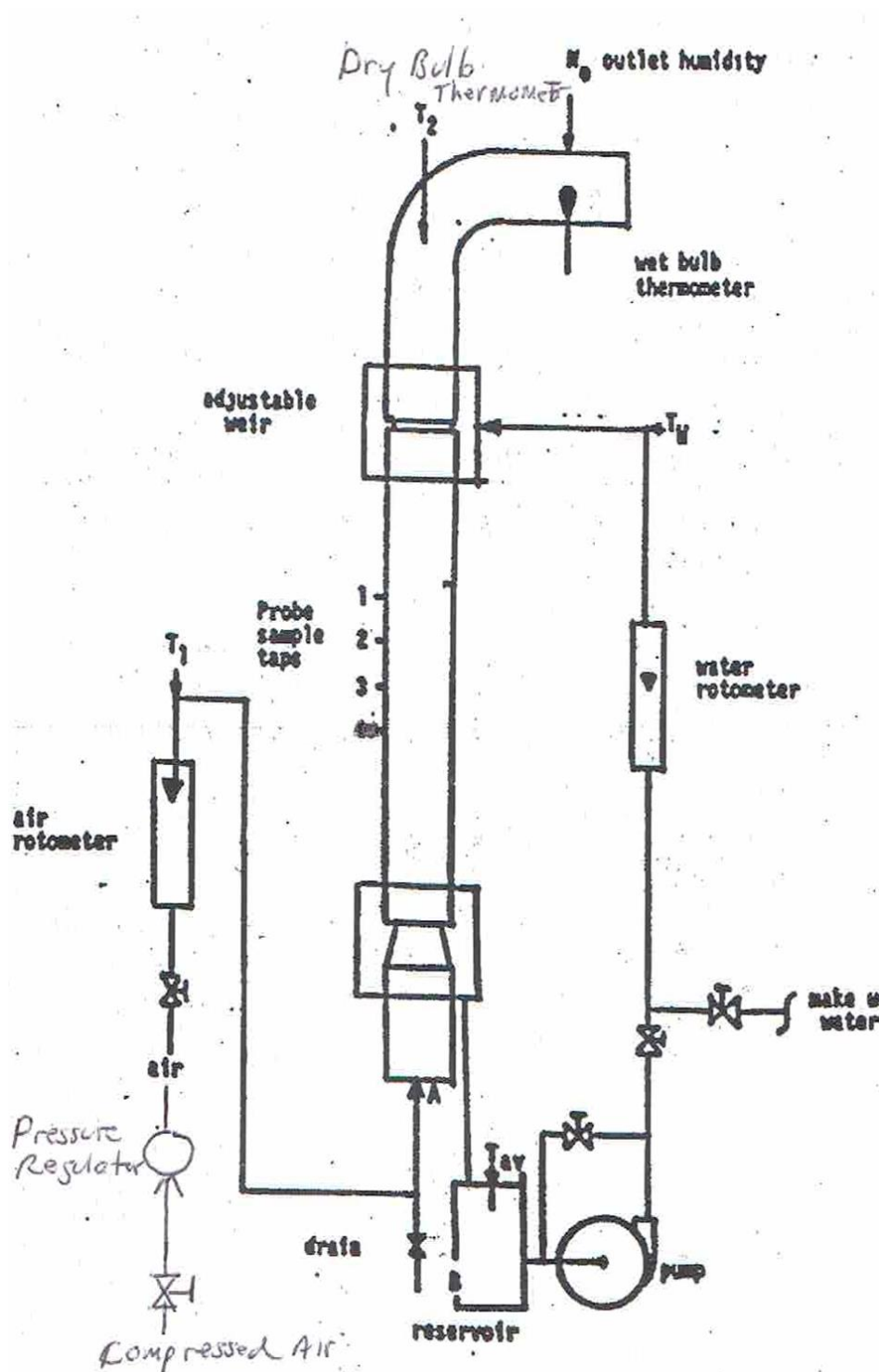


Figure 3. Schematic of the Unit Operation Wetted Wall Column.

Experiment Methods. It is necessary to select a series of experiments involving the variables of:

1. Liquid flowrate
2. Air flowrate
3. Temperature of the incoming air

The choice and number of these experiments will be dictated by how the experimental data will be analyzed.

Before beginning the experiment, measure the humidity of the incoming air by using the wet and dry bulb thermometers or the electronic hygrometer. Also measure the air temperature. This establishes the humidity of the incoming air.

The experiment is started by circulating water in the column at a sufficient rate to completely wet the inside wall of the tube. Next, establish an air flow (unheated) and allow the circulating water temperature to reach steady state. Measure the steady state values of the water and air temperatures at the various locations in the column as well as the air humidity using the wet and dry bulb thermometers or the electronic hygrometer. From the data, the values of k_y , h and N_{tG} can be calculated using equations X and Y.

Repeat the experiment for two other air flowrates. Then, repeat these same three air flowrates using heated air. The incoming air can be heated to two different temperatures using the heat exchanger which uses steam on the shell-side to heat the air. A psychrometric chart for the air-water system can be used to aid in the calculations.

Data Analysis

The data can be reduced to give heat and mass transfer coefficients. Plot h and k_y as a function of the Reynolds number. Also, plot the Sherwood number divided by $Sc^{0.44}$ as a function of the Reynolds number on a $\log_{10}$ - $\log_{10}$ plot. How does your measured data compare to the correlation given by equation 3?

Determination of these values allows a comparison with existing correlations for air-water systems in wetted-wall columns. The goal should be a qualitative study of the expected effect of the system variables on the transport coefficients and also a quantitative comparison. If the agreement is not good, an analysis should be presented on the suspected experimental problems and how the experimental procedure could be improved.

The data can also be reduced to determine of HTU and NTU values. These can also be analyzed as a function of system parameters.

Questions and Further Analysis to Consider

1. Plot NTU as a function of the inlet air temperature with air flowrate as a parameter. How do you expect the transport coefficients to depend on the air inlet temperature? Does the water flowrate effect the transport coefficients?
2. Compare your measured values of k_Y to values obtained from the correlation given by equation (3).
3. Show, by making an overall mass and energy balance on the column, that $h/(k_Y C_s)=1$. This is known as the Lewis relationship.

References

1. Treybal, R.E. "Mass Transfer Operations", 3rd Edition, McGraw-Hill, New York (1980).
2. McCabe, W.L., Smith, J.C., Harriott, "Unit Operations in Chemical Engineering" 7th Edition, McGraw-Hill, New York.
3. Bird, R.B., Stewart, W.E., Lightfoot, E., "Transport Phenomena", Wiley, New York.
4. Perry's Chemical Engineering Handbook, ...

List of Symbols (SI Units)

Note that the symbols, units and definitions used in this manual are self-consistent. However, be careful when referring to another reference or textbook since they may use a different set of conventions. You should look at the concepts discussed here and in your textbook and reconcile any differences in definitions.

A	Area for mass or heat transfer	(m ²)
c	molar concentration	(kmol/m ³)
c _A	molar concentration of component A	(kmol/m ³)
C _p	molar heat capacity	(J/kmol - K)
C' _A	heat capacity of vapor	(J/kg - K)
C' _B	heat capacity of gas	(J/kg - K)
C' _s	heat capacity of gas & vapor	(J/kg - K)
d _p	column diameter	(m)
D _{AB}	diffusion coefficient	(m ² /s)
G	molar flowrate of gas	(kmol/s)
G' _s	mass flowrate of dry gas	(kg/s)
h	heat transfer coefficient	(J/m ² - s - K)
k _c	mass transfer coefficient	(m/s)
k _Y	mass transfer coefficient	(kg/m ² - s)
MW	molecular weight	(kg/kmol)
N _A	molar flux of A	(kmol/m ² - s)
N _{tG}	number of gas phase transfer units	(dimensionless)
P	column perimeter	(m)
q	heat flux	(J/m ² - s)
Re	Reynolds number	(ρd _p u _o /μ)
Sc	Schmidt number	μ/ρD _{AB}
Sh	Sherwood number	
y	mole fraction	
Y	mass ratio	